

- Naming convention: Practical1_Scan_RollNo.pdf, Practical1_Output_RollNo.png, VivaAnswers_RollNo.pdf
- Plagiarism in handwriting or viva answers will result in cancellation of the entire assignment. Write in your own words.
- Deadline is strictly enforced. Late submissions will not be accepted unless prior permission is obtained.
- For any queries, post a comment on Google Classroom — do not message personally.

Practical List & Marks Distribution

No.	Title	Topic / Unit	Marks
1	Array Operations – Insertion, Deletion & Traversal	Arrays – Unit 1	5
2	Searching – Linear Search & Binary Search	Arrays / Searching – Unit 1	5
3	Sorting – Bubble Sort & Selection Sort	Arrays / Sorting – Unit 1	5
4	Singly Linked List – Create, Insert & Delete	Linked Lists – Unit 2	5
5	Doubly Linked List – Insert & Delete	Linked Lists – Unit 2	5
6	Stack Implementation Using Array	Stacks – Unit 2	5
7	Stack Application – Parenthesis Matching	Stacks / Applications – Unit 2	5
8	Queue Implementation Using Array	Queues – Unit 2	5
9	Circular Queue Implementation	Queues – Unit 2	5

```
        (*n)--;\n    }\n    int main() {\n        int a[MAX], n;\n        printf("Enter n: "); scanf("%d",&n);\n        for(int i=0;i<n;i++) scanf("%d",&a[i]);\n        display(a,n);\n        // Insert and delete as per user input\n        return 0;\n    }
```

Expected Output

Show array before insertion, after insertion, and after deletion.

Viva Questions

- Q1.** What is the time complexity of insertion at the beginning of an array?
- Q2.** Why does deletion at position 0 require shifting all elements?

Practical 3: Sorting – Bubble Sort & Selection Sort

Unit: Unit 1 | Topic: Arrays / Sorting

Aim

Write a C program to sort an array using Bubble Sort and Selection Sort techniques.

Objectives

- Implement Bubble Sort and observe the pass-by-pass sorting process.
- Implement Selection Sort and compare it with Bubble Sort.
- Print the array after each pass to understand the algorithm.

Steps to Follow

11. Accept an array of n integers from the user.
12. Sort using Bubble Sort and display array after each pass.
13. Reset the array and sort using Selection Sort.
14. Compare the total number of swaps in both methods.

Reference Code (Study & Write in Notebook by Hand)

Note: Do NOT copy-paste. Understand the logic and write the complete program in your own handwriting.

```
#include<stdio.h>
void bubbleSort(int a[], int n) {
    for(int i=0; i<n-1; i++)
        for(int j=0; j<n-i-1; j++)
            if(a[j]>a[j+1]) { int t=a[j]; a[j]=a[j+1]; a[j+1]=t; }
}
void selectionSort(int a[], int n) {
    for(int i=0; i<n-1; i++) {
        int min=i;
        for(int j=i+1; j<n; j++)
            if(a[j]<a[min]) min=j;
        int t=a[i]; a[i]=a[min]; a[min]=t;
    }
}
```

Expected Output

Sorted array and number of swaps for each sorting method.

Viva Questions

- Q1.** What is the time complexity of Bubble Sort in the worst case?
- Q2.** Which sorting algorithm is better in terms of number of swaps — Bubble or Selection?

Practical 5: Doubly Linked List – Insert & Delete

Unit: Unit 2 | Topic: Linked Lists

Aim

Implement a doubly linked list in C and perform insertion and deletion from both ends.

Objectives

- Understand the structure of a doubly linked list with prev and next pointers.
- Insert nodes at the beginning and end of the list.
- Delete nodes from the beginning and end.
- Traverse the list in both forward and backward directions.

Steps to Follow

20. Define a node with data, prev, and next pointers.
21. Implement insertAtBeginning() and insertAtEnd().
22. Implement deleteFromBeginning() and deleteFromEnd().
23. Implement forwardTraversal() and backwardTraversal().
24. Demonstrate with at least 4 nodes.

Reference Code (Study & Write in Notebook by Hand)

Note: Do NOT copy-paste. Understand the logic and write the complete program in your own handwriting.

```
#include<stdio.h>
#include<stdlib.h>
struct Node { int data; struct Node *prev, *next; };
struct Node* createNode(int val) {
    struct Node* n = malloc(sizeof(struct Node));
    n->data=val; n->prev=n->next=NULL;
    return n;
}
```

Expected Output

Forward and backward traversal of the list after each operation.

Viva Questions

- Q1.** How does a doubly linked list differ from a singly linked list in terms of memory?
- Q2.** What is the time complexity of inserting a node at the beginning of a doubly linked list?